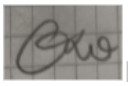
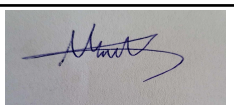
EE3014 Energy Conversion

Experiment 6

Student No	Name	Surname	Contribution	Signature
150722923	Jacques	Asinyo	Abstract, Introduction, theoretical background	
150725824	Kadir Baha	Demirhan	Results, Conclusion	
150720062	Mehmet Ali	KAMALAK	Procedure and experimental setup	
150722066	Mevlüt	Akhan	Experimental Setup, Procedure and Graphs	

Abstract

This report presents the experimental investigation of a DC series motor's fundamental operating characteristics. The primary objectives were to analyze the relationship between the motor's terminal voltage and its speed, and to determine the speed-torque performance under varying mechanical loads. A dynamometer braking system was utilized to apply a precisely controlled and measurable load torque. The first experiment involved varying the terminal voltage while maintaining a constant load, and the second involved varying the load torque while maintaining a constant terminal voltage. The results demonstrate that the motor's speed is highly dependent on both voltage and load. Crucially, the speed-torque analysis revealed a characteristic inverse relationship, with the motor's speed dropping sharply as load torque increases. This confirms the DC series motor's defining features: extremely high starting torque and a variable speed profile, which are in strong agreement with established theoretical principles.

1. Introduction

1.1. Background and Motivation

The DC series motor is a cornerstone of high-torque electric drive applications, historically used in everything from electric trains to industrial cranes. Unlike other motor types, its unique configuration, where the field winding is in series with the armature, gives it a distinct set of performance characteristics. Understanding these characteristics (specifically, how its speed responds to changes in voltage and load) is essential for any engineer working with electric machinery. This experiment aims to provide a practical and hands-on validation of these theoretical principles.

1.2. Objectives of the Experiment

The primary objectives of this laboratory exercise are:

1. To experimentally determine the relationship between the terminal voltage and the rotational speed of a DC series motor under a constant load.
2. To experimentally determine the speed-torque and speed-current characteristics of the DC series motor under a constant supply voltage.
3. To compare the observed performance curves with the theoretical models and understand the practical implications of these characteristics.

1.3. Scope of the Report

This report begins with a detailed theoretical background on the DC series motor, including a comparison with the DC shunt motor and an explanation of the dynamometer braking system used. It then outlines the experimental procedure, followed by a presentation and analysis of the collected data in graphical form. The report concludes with a discussion of the results and their alignment with theory.

2. Theoretical Background

2.1. The DC Series Motor

A DC series motor is characterized by its field winding being connected in series with the armature winding, as shown in the equivalent circuit in **Figure 1**.

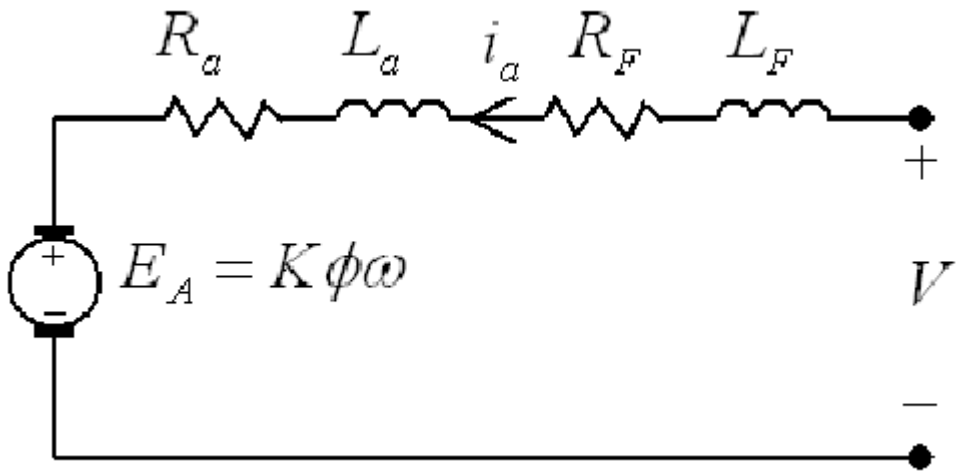


Figure 1: DC Series Motor Equivalent Circuit. [1]

This series connection is the most important feature. It means the armature current (I_a), the field current (I_f), and the total line current (I_L) are all identical:
I_a = I_f = I_L

The magnetic flux (Φ) produced by the field winding is directly proportional to the field current (Φ ∝ I_f). Since I_f = I_a, the flux is directly proportional to the armature current:
Φ ∝ I_a

This relationship governs the motor's unique performance:

- **Torque:** The general torque equation is T ∝ Φ * I_a. Substituting our flux relationship gives:
  T ∝ (I_a) * I_a => **T ∝ I_a²**
  This squared relationship means that at low speeds (and high currents, like during startup), the DC series motor produces an exceptionally high torque, making it ideal for starting heavy loads.
- **Speed:** The general speed equation is N ∝ E_A / Φ. Since the back EMF (E_A) is approximately equal to the terminal voltage (E_A ≈ V_T), we have:
  N ∝ V_T / I_a
  This shows two things:
  1. The speed is inversely proportional to the load current (I_a). As the load increases, the motor automatically slows down.

2. At no-load, I_a is extremely small, causing the speed N to become dangerously high. For this reason, **a series motor must never be run without a load.**

2.2. The DC Shunt Motor (For Comparison)

To appreciate the series motor, it is useful to contrast it with a DC shunt motor, where the field winding is connected in parallel with the armature (**Figure 2**).

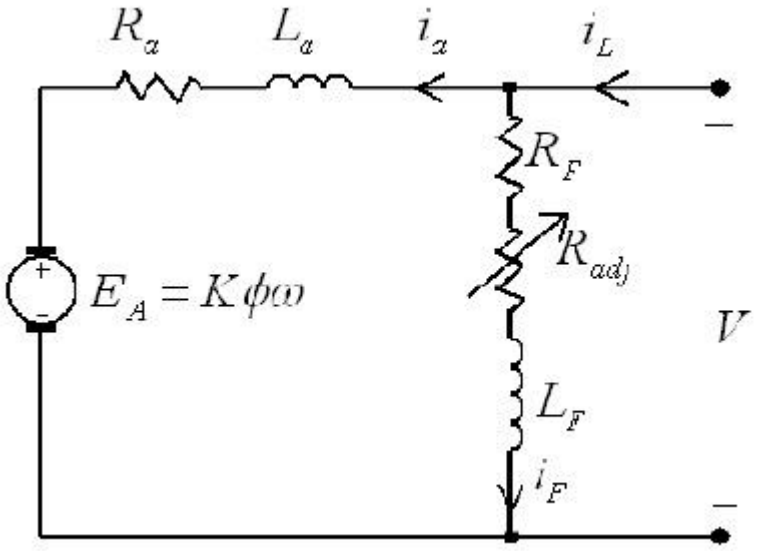


Figure 2: DC Shunt Motor Equivalent Circuit. [2]

In a shunt motor, the field winding receives a constant terminal voltage, resulting in a constant field current and therefore a **nearly constant magnetic flux (Φ)**. This leads to very different characteristics:

- **Torque:** Since Φ is constant, T ∝ I_a (a linear relationship).
- **Speed:** Since Φ is constant, N ∝ E_A. The speed remains relatively constant from no-load to full-load, making it suitable for constant-speed applications like fans and machine tools.

2.3. The Dynamometer Braking System

In this experiment, the mechanical load was applied using a dynamometer. A dynamometer is a device that measures torque and speed to determine the mechanical power of an engine or motor. The type used in this lab is likely an eddy-current brake. It operates as follows:

1. A metal disc (rotor) is coupled to the shaft of the motor under test.
2. An electromagnet, controlled by an external DC power supply, creates a magnetic field that passes through the disc. The "load voltage" you changed in Part 2 was the control voltage for this electromagnet.
3. As the motor spins the disc through the magnetic field, eddy currents are induced in the disc.

4. According to Lenz's Law, these eddy currents create their own magnetic field that opposes the main field, resulting in a braking torque that resists the motor's rotation.
5. By increasing the "load voltage," we increase the electromagnet's strength, which induces larger eddy currents and produces a stronger braking torque. A load cell measures this reaction torque, which is displayed on the control unit.

3. Experimental Setup and Procedure

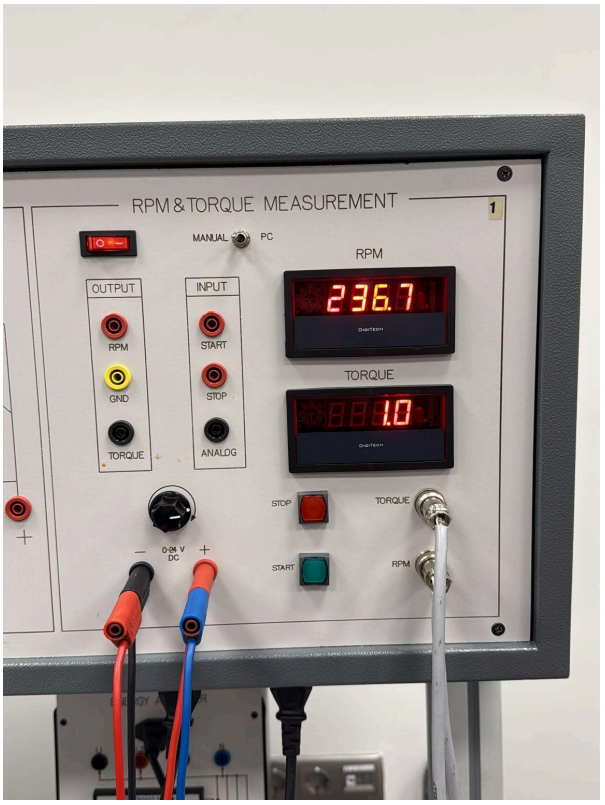
3.1 Equipment Used

- DC Series Motor
-
- Multimeter
- Connection Cables

3.2 Part 1 – Voltage vs RPM Test Setup



The braking system is set at 2.0 Volts for safety work.



We started at 30 Volts to measure RPM.



For 52.5 Volts measured RPM value



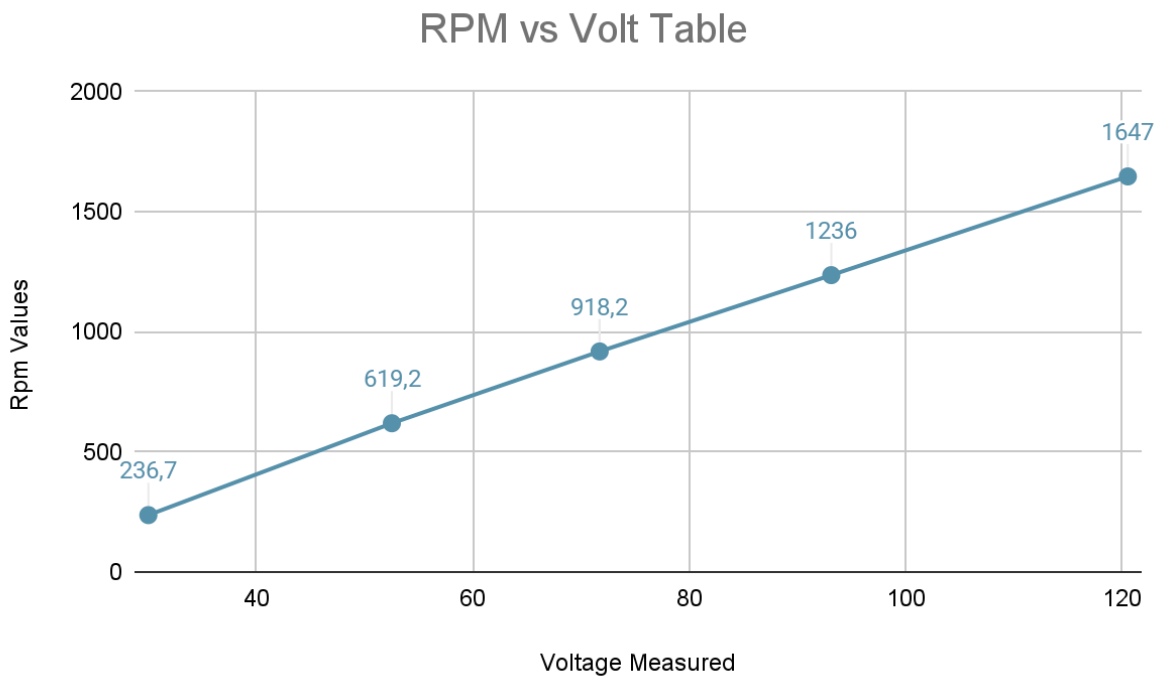
For 71.7 Volts measured RPM value



For 93.1 Volts measured RPM value



For 120.5 Volts measured RPM value

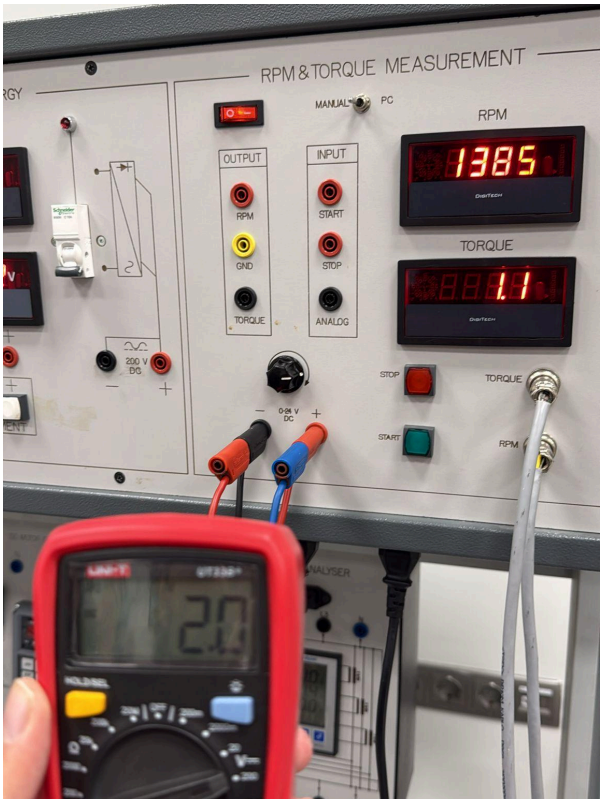


3.2 Part 1 – Voltage vs RPM Test Setup Procedure

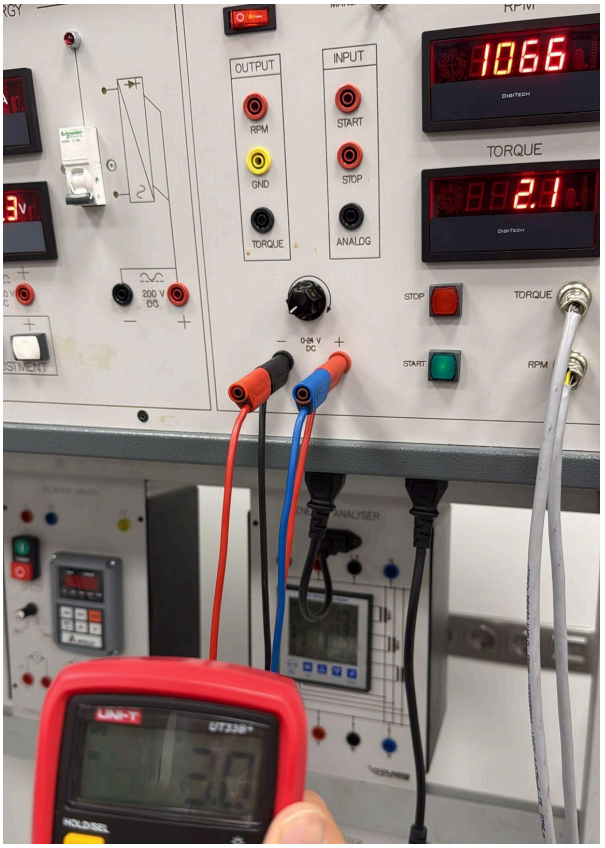
In the first part of the experiment, the braking system was initially adjusted to a control voltage of 2.0 V to ensure safe operation. After verifying all electrical connections, the motor setup was energized and the power supply was kept at a low initial voltage. The terminal voltage was gradually increased, allowing the motor to reach steady-state operation at each step, and the corresponding speed values were read from the measurement unit. Measurements were taken starting from 30 V and subsequently at 52.5 V, 71.7 V, 93.1 V, and 120.5 V. During this process, the mechanical load remained constant, and only the terminal voltage was varied in order to investigate how the motor speed responds to changes in supply voltage. The collected data were later used to construct the voltage–speed characteristic curve of the DC series motor.

3.2 Part 2 – Load Torque vs Speed

The supply voltage was fixed at 120 V and the brake torque was subsequently increased to record the speed–torque characteristic.



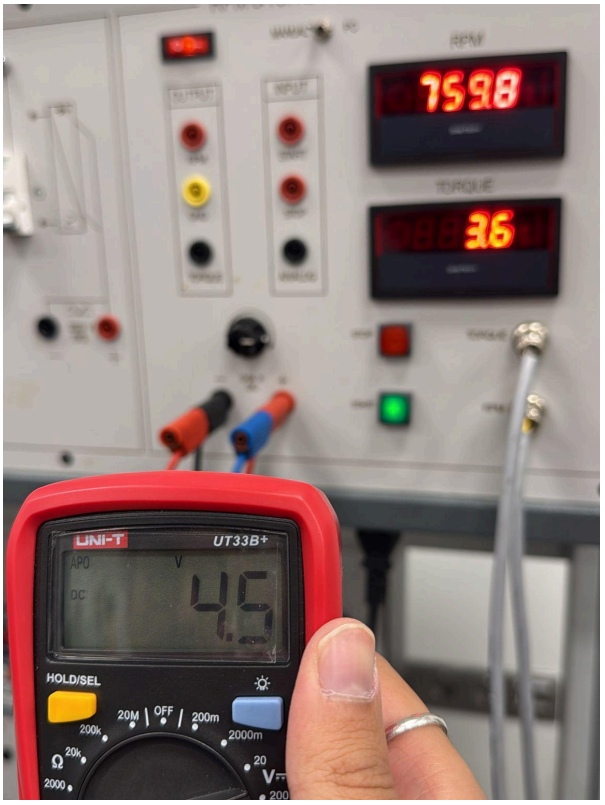
At 2 Volt brake measurement



At 3 Volt brake measurement



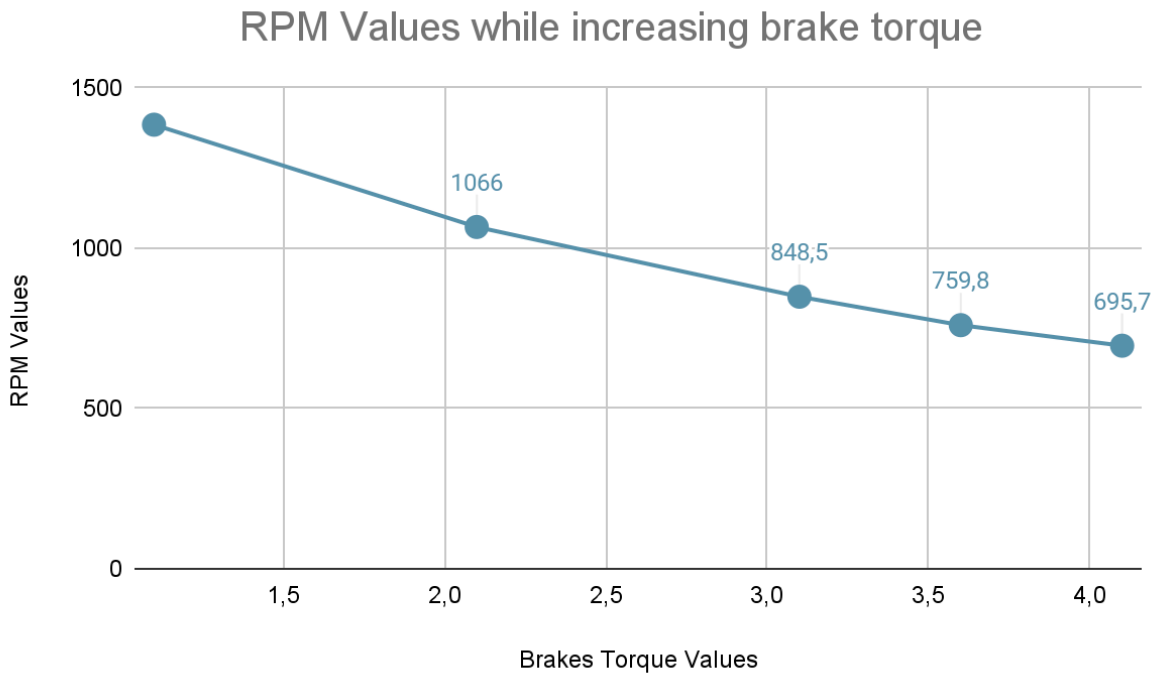
At 4 Volt brake measurement



At 4.5 Volt brake measurement



At 5 Volt brake measurement



3.2 Part 2 – Load Torque vs Speed Procedure

In the second part of the experiment, the supply voltage was fixed at 120 V in order to examine the motor's speed response under varying load torque. Once the motor reached stable operation, the control voltage feeding the dynamometer's excitation coil was gradually increased. Increasing this control voltage strengthened the eddy current braking effect, thereby applying a higher mechanical load torque to the motor shaft. At each braking voltage level, the motor speed was recorded from the measurement panel. Readings were taken at 2 V, 3 V, 4 V, 4.5 V, and 5 V, allowing observation of the progressive decrease in motor speed as the applied load torque increased. This procedure enabled the experimental determination of the characteristic speed–torque behavior of the DC series motor.

4.Results

This section summarizes the experimental results from two key tests on the DC series motor: (1) the voltage–speed relationship under a constant load, and (2) the speed–torque behavior under slowly increasing mechanical load.

4.1 Voltage vs. Speed (RPM) Results

In the first part of the experiment, the braking system was held constant at 2.0 V, while the motor's terminal voltage was adjusted across a range. At each voltage setting, the motor was allowed to reach steady-state conditions before its speed was recorded from the measurement panel.

The measurements are as follows:

Terminal Voltage (V)	**Speed (RPM)**
30.0	236.7
52.5	619.2
71.7	918.2
93.1	1236
120.5	1647

The results show that as the terminal voltage increases, the motor speed rises almost linearly. This trend, clearly visible in the "RPM vs Volt Table" graph, matches what we expect from theory. For a DC series motor under light load, the back EMF grows in step with the applied voltage, which leads to higher speed.

4.2 Load Torque vs. Speed Results

During the second part of the experiment, the supply voltage remained fixed at 120 V. To increase the mechanical load, we raised the excitation voltage applied to the dynamometer brake. Each step-up in braking voltage translated directly to a higher load torque.

The measurements are as follows:

Brake Voltage (V)	Speed (RPM)	Load Torque
2.0	1385	1.1
3.0	1066	2.1
4.0	848.5	3.1
4.5	759.8	3.6
5.0	695.7	4.1

The graph titled “RPM Values while increasing brake torque” clearly shows that as brake torque increases, motor speed steadily decreases. This is a well-known trait of DC series motors. When the load becomes heavier, the armature current rises. That increase strengthens the magnetic field, which naturally slows the motor down.

5. Conclusion

This experiment clearly showed how a DC series motor reacts under different conditions.

In the first part, we changed the terminal voltage. When the voltage was low, the motor turned slowly. As the voltage increased, the speed rose almost in a straight-line pattern. The faster the voltage climbed, the faster the motor spun.

In the second part, we kept the voltage steady but increased the mechanical load. As the load got heavier, the motor slowed down. This happened in a smooth and expected way. DC series motors slow down under heavier loads because they need more current to keep turning.

In both tests, the motor acts just as theory predicts: it speeds up with more voltage and slows down when more torque is needed. That's why these motors are great for situations where strong starting power and adaptable speed are important.

References

[1]
https://www.researchgate.net/figure/Equivalent-Circuit-of-DC-Series-Motor_fig2_269484406

[2]
https://www.researchgate.net/figure/Equivalent-Circuit-of-DC-Series-Motor_fig2_269484406